NEXTBUY

From raw data to smart decisions

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Notre approche

01 Etude du sujet et mise en place des questions à traiter

→ notebook

02 Mise en place des imports au début du notebook et réponse aux questions via des recherches dont les résultats sont présentés sous forme de graphiques.

03 Feature engineering et machine learning, puis compression et exportation des modèles

04 Tests unitaires et formatage via Ruff

Starting point

We started by writing the 8 questions we will be using to obtain business insight based on the data provided.

1 - Which products are the bestsellers?

2 - What other products do users frequently order with chocolate ?

3 - What aisle and departments have the best selling rate ?

4 - What are the main customer profiles, such as midnight shopper, casual buyer, or cart addict ?

5 - Is there a good timing to send an offer to someone based on the last time he ordered ?

6 - What products should we recommend to a customer ?

7 - Can we predict whether a customer will reorder a given product in their next order ?

8 - Can we predict how many days after their last observed order a customer will purchase a given product again ?

First we make all the import we will need in one place to keep the notebook clean and will be adding libraries in this bloc if need be

```
import squarify
import joblib
import pandas as pd
from copy import deepcopy
import matplotlib.pyplot as plt
from wordcloud import WordCloud
from sklearn.model_selection import train_test_split, GridSearchCV, ParameterGrid, RandomizedSearchCV
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
from sklearn.dummy import DummyClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.pipeline import Pipeline
from sklearn.linear_model import HuberRegressor, LinearRegression, Ridge, ElasticNet
from sklearn.dummy import DummyRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor, ExtraTreesRegressor, GradientBoostingRegressor, AdaBoostRegressor
from sklearn.neighbors import KNeighborsRegressor
```

Tests et formatage

```python
[tool.ruff]
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target-version = "py314"
exclude = [
    ".git",
    ".venv",
    "venv",
    "__pycache__",
]

[tool.ruff.lint]
select = [
    "E",   # pycodestyle errors
    "F",   # pyflakes
    "W",   # pycodestyle warnings
    "I",   # isort
    "UP",  # pyupgrade
    "B",   # flake8-bugbear
    "SIM", # flake8-simplify
    "C90", # mccabe
]
ignore = [
    "E501",   # auto line length
]

fixable = ["ALL"]
unfixable = []

[tool.ruff.lint.mccabe]
max-complexity = 10

[tool.ruff.lint.per-file-ignores]
"*.ipynb" = ["E402", "F401"] # imports
"tests/*.py" = ["S101"] # asset

[tool.ruff.format]
quote-style = "single"
indent-style = "space"
line-ending = "auto"
docstring-code-format = true
```
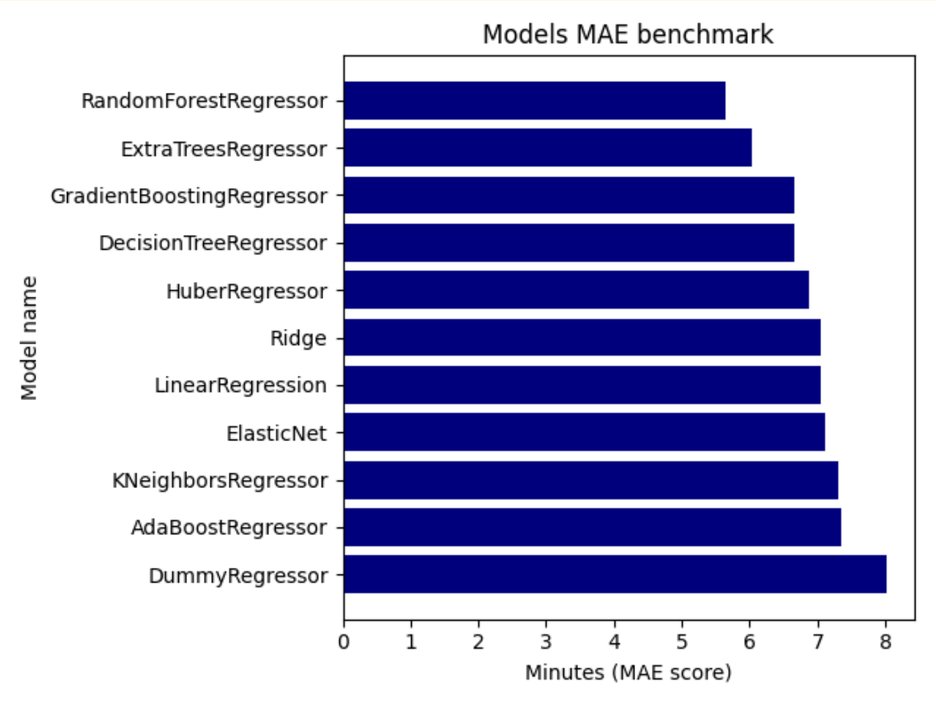
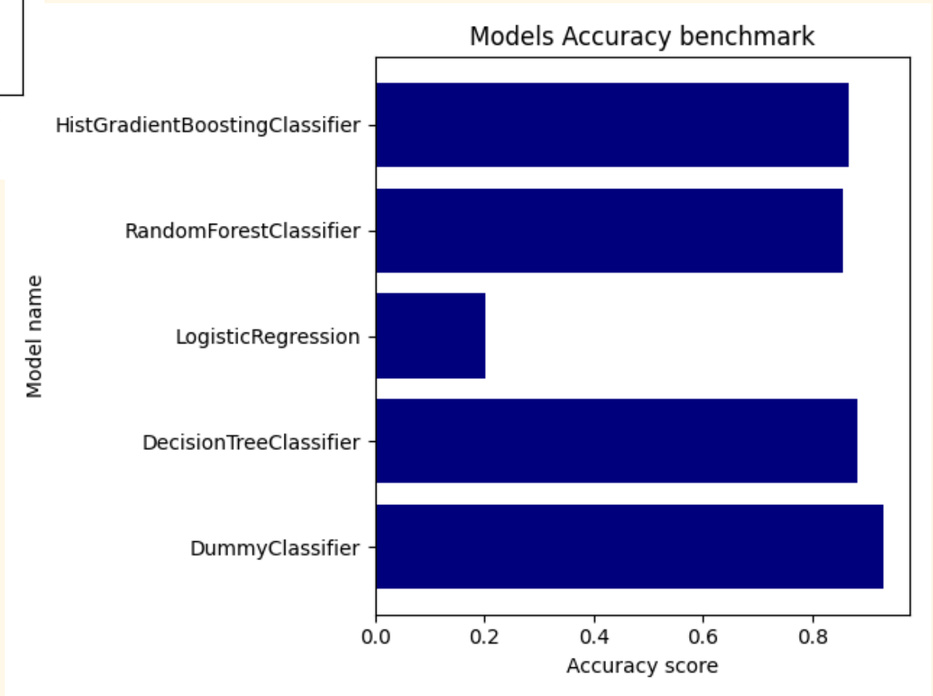

```python
from testbook import testbook

@testbook(
    'notebook.ipynb',
    execute=['imports', 'load_datasets', 'merge_datasets', 'get_best_sellers'],
)
def test_get_best_sellers_type(tb):
    func = tb.get('get_best_sellers')
    result = func()
    assert hasattr(result, 'isna')

@testbook(
    'notebook.ipynb',
    execute=['imports', 'load_datasets', 'merge_datasets', 'aisles_dep_selling_rate'],
)
def test_aisles_dep_selling_rate(tb):
    func = tb.get('aisles_dep_selling_rate')

    result_dep = func('department', 5)
    print(result_dep.shape)
    assert hasattr(result_dep, 'isna') and result_dep.shape[0] == 5

    func('aisle', 5)
    print(result_dep.shape)
    assert hasattr(result_dep, 'isna') and result_dep.shape[0] == 5
```

Les modèles

Dans le cadre de la réponse à 2 questions nous avons réalisés plusieurs benchmark sur 11 modèles.



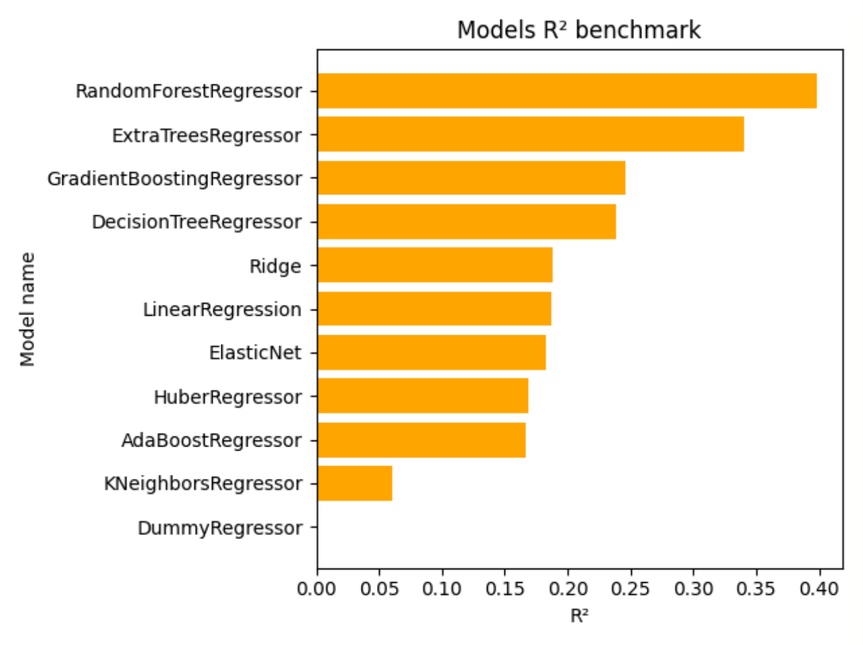


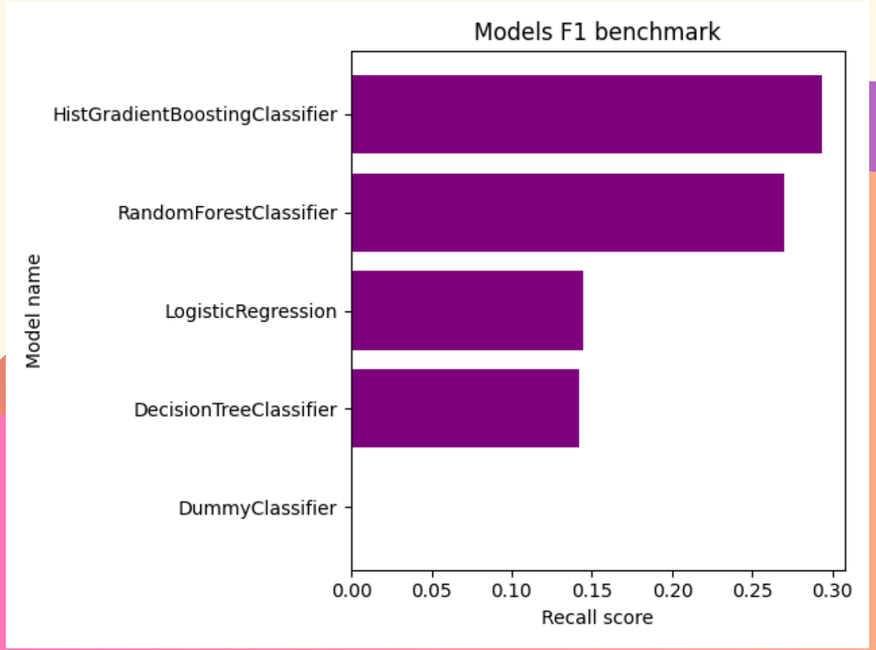
Les modèles choisis

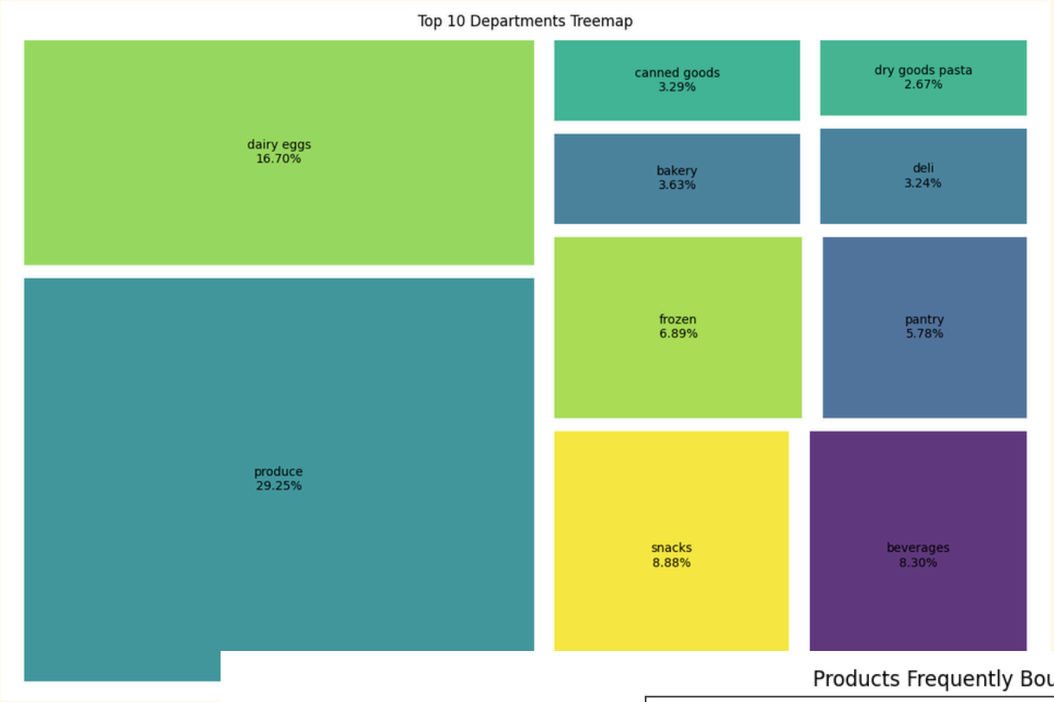
Le premier modèle choisi est “HistGradientBoostingClassifier”, le second est “RandomForestRegressor”.

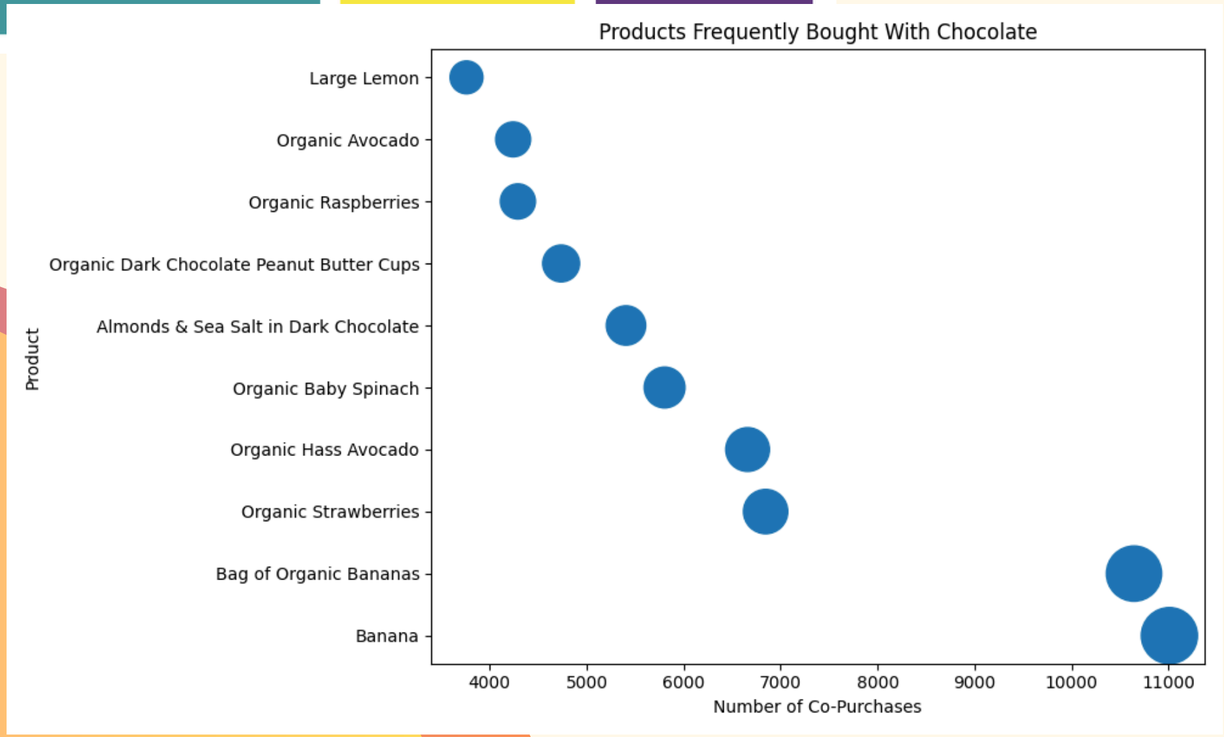
Chaque modèle répondant à un besoin spécifique il est logique d’obtenir une réponse différente.

Pour chaque modèles nous avons utilisés plusieurs benchmarks que nous détaillons dans le notebook.







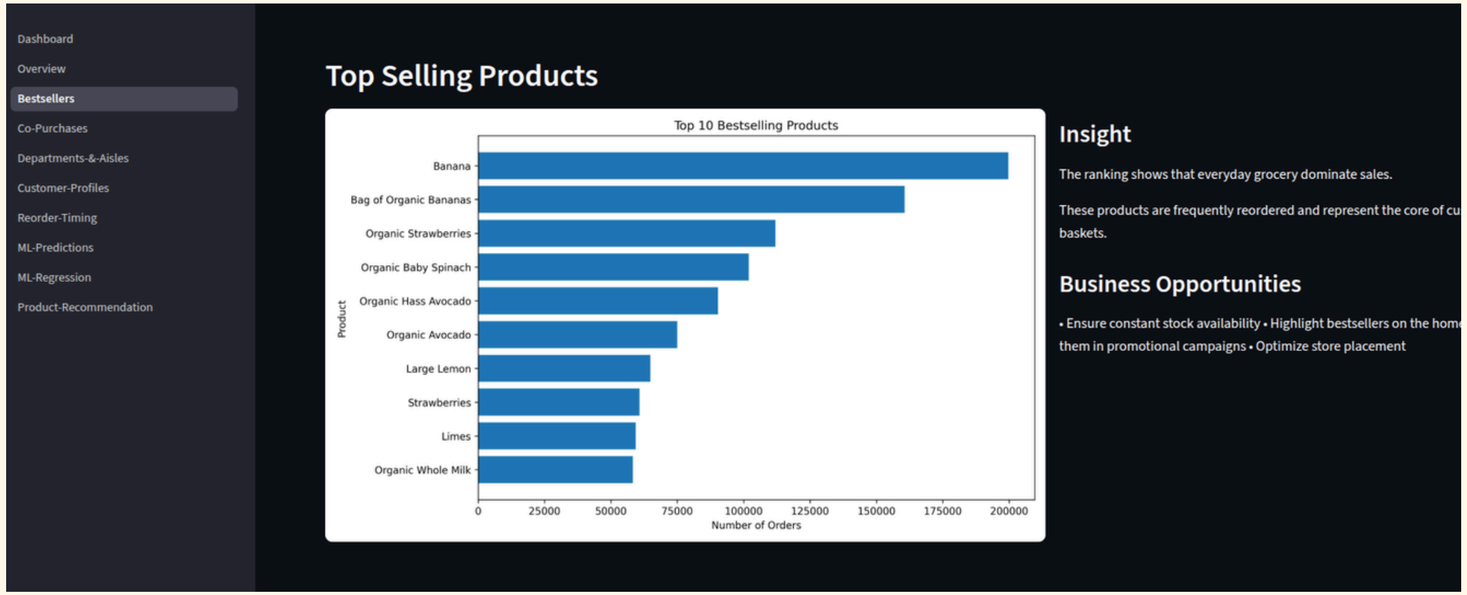


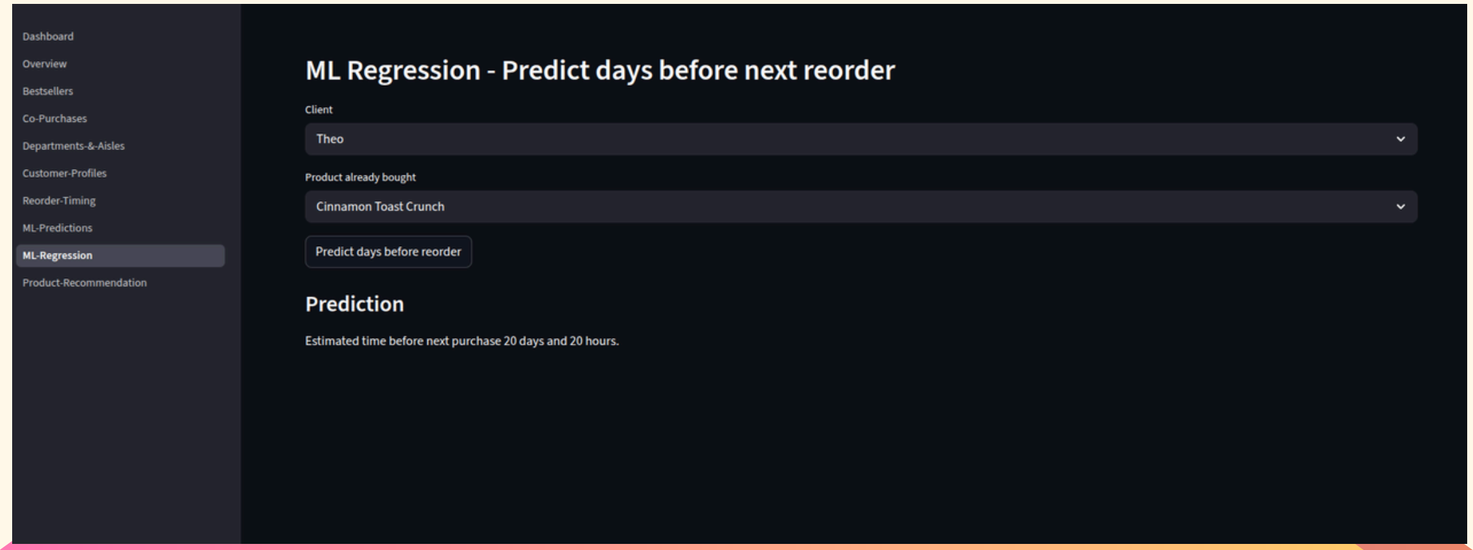
Les graphiques

Nextbuy possède beaucoup de graphiques :

- Chacun aide à visualiser un assemblage de données choisis avec soin pour en tirer des informations utiles à la compagnie.
- Chaque graphique est accompagné d'une explication et d'une conclusion pouvant être immediatement utilisée pour maximiser les ventes ou pour mieux comprendre les habitudes d'achat des clients.
- Chacun d'entre eux est unique et adapté au type de données étudié.

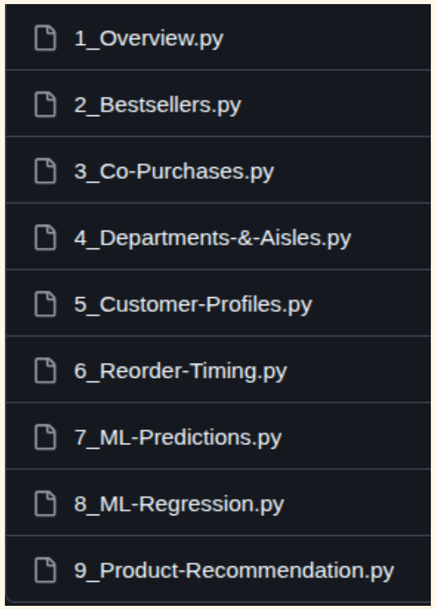
Le Dashboard





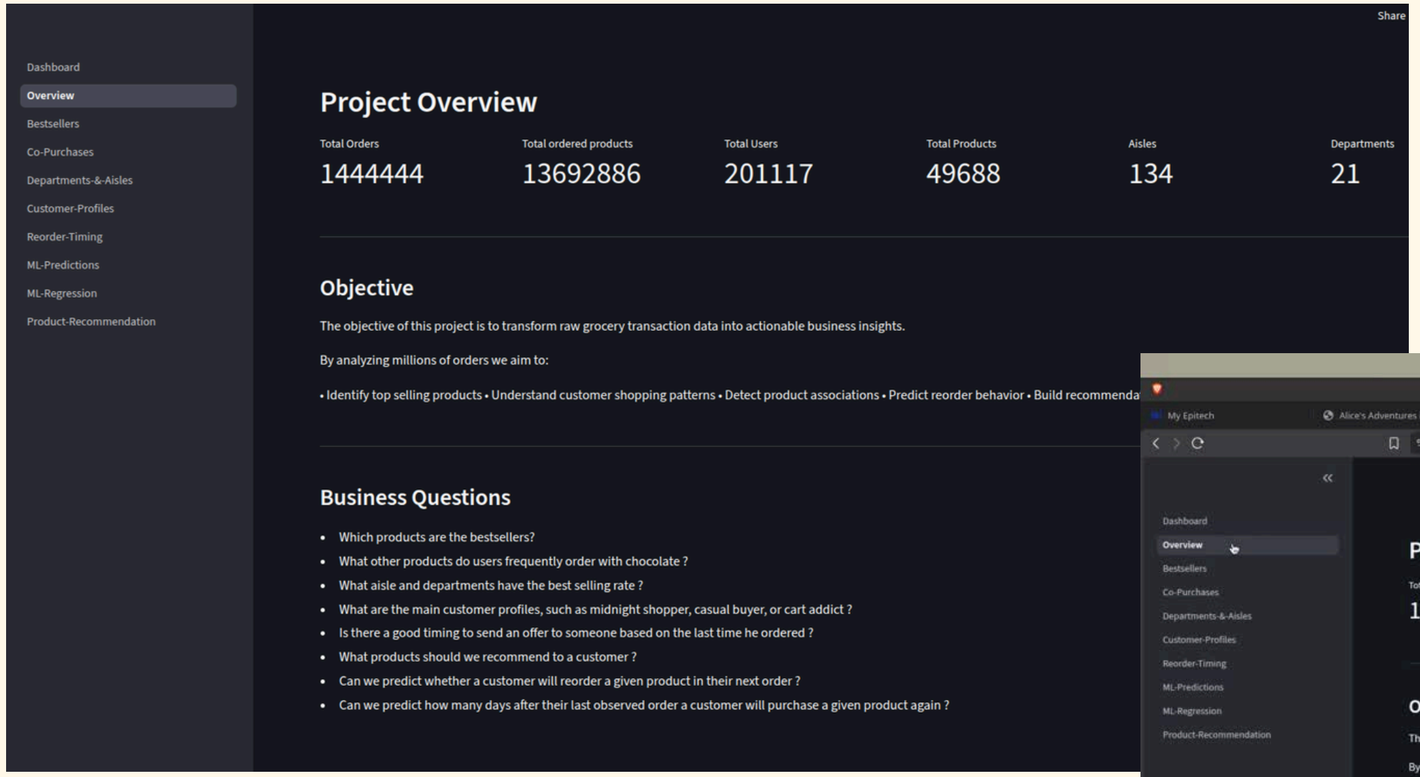
Nous avons décidés de présentés nos résultats sous la forme d'un dashboard constitués de plusieurs pages:

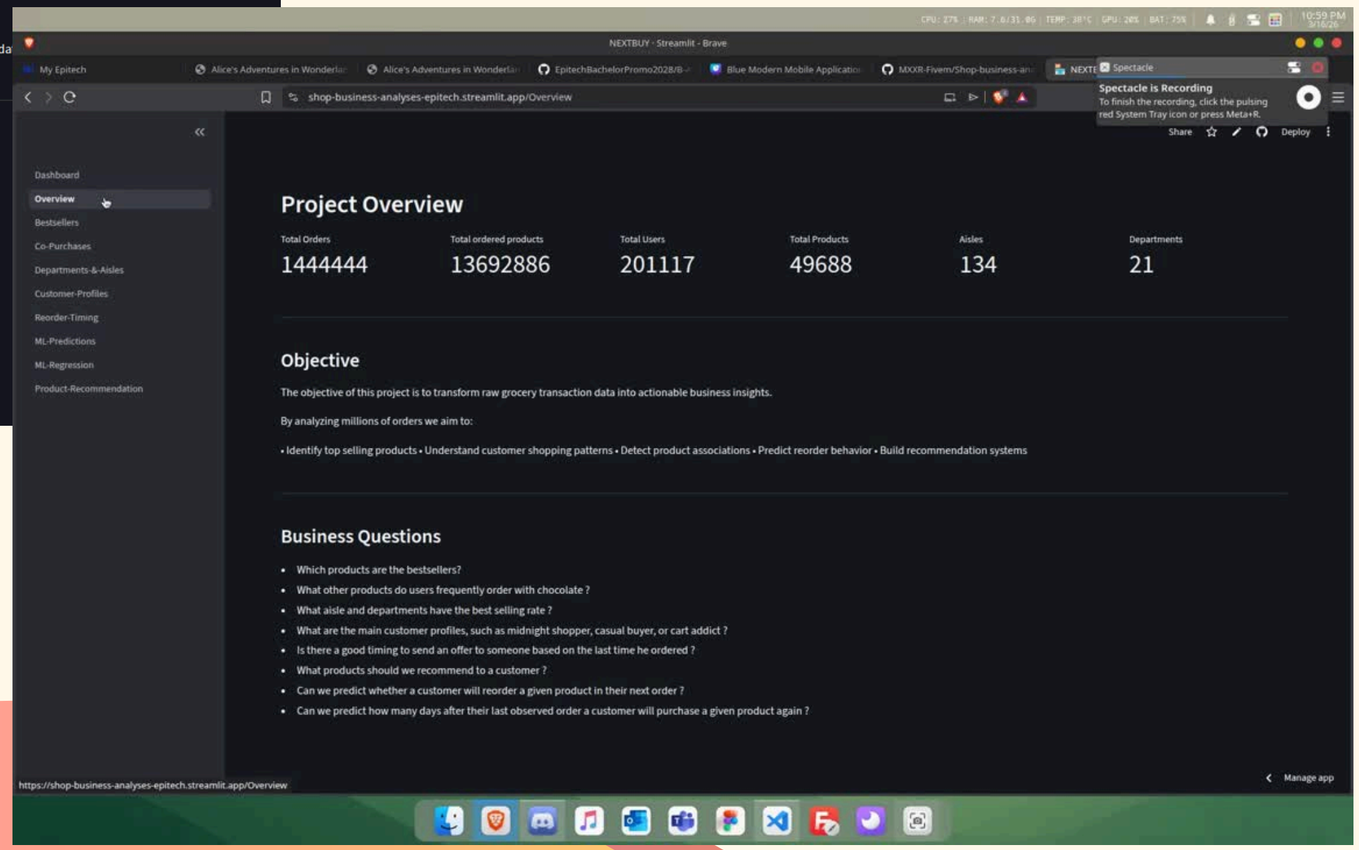
- La première partie est un ensemble de graphique presentés et expliqués pour en tirer des leviers utiles
- La seconde partie contient 3 modules permettant de faire des prédictions utiles basées sur un input utilisateur, permettant de voir visuellement ce que les modèles peuvent faire à grande échelle pour aider le business à intégrer des promotions et interpréter des données.



L’hébergement

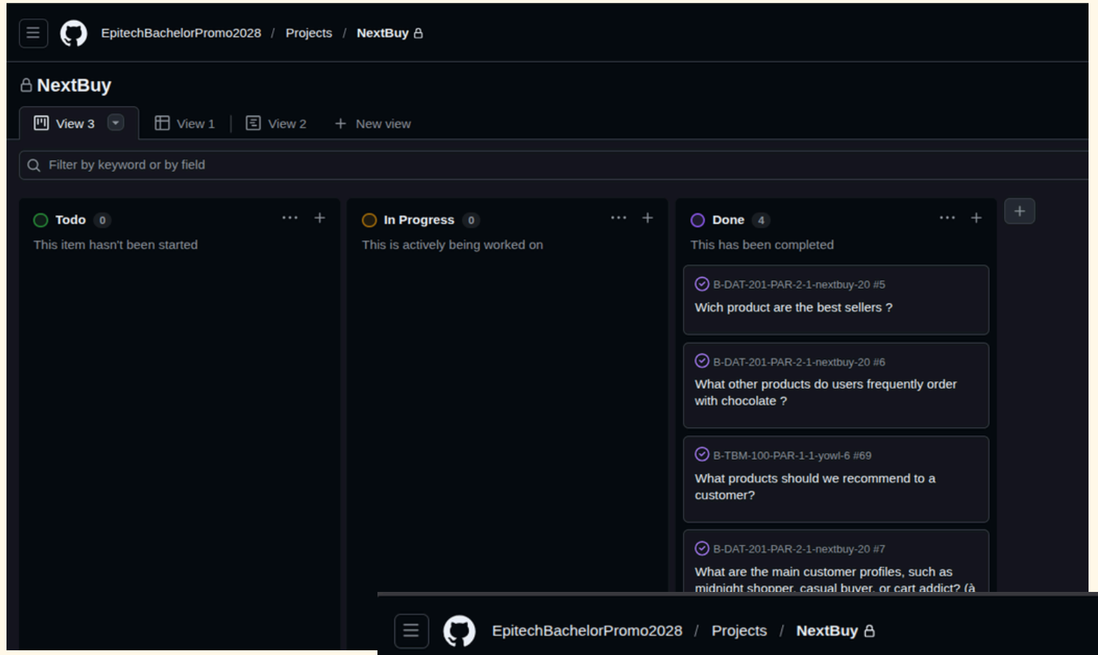
Nextbuy est hébergé sur l’outil Streamlit community qui permet un hébergement simple et rapide des applications développées sur Streamlit.





Les outils

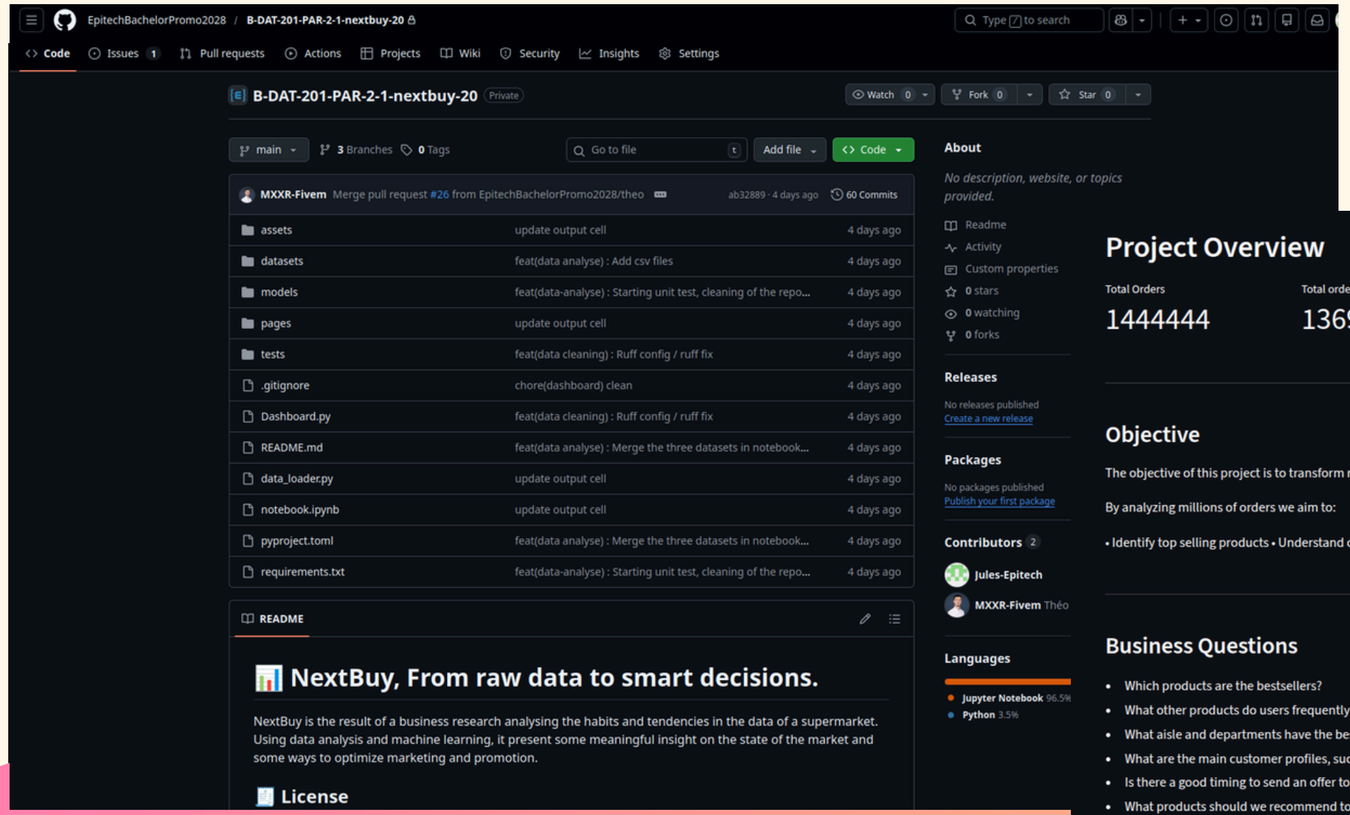
Pour le développement nous avons utilisé Notion pour la première partie, definition des périodes et du workflow et github project pour établir une todo claire et précise.

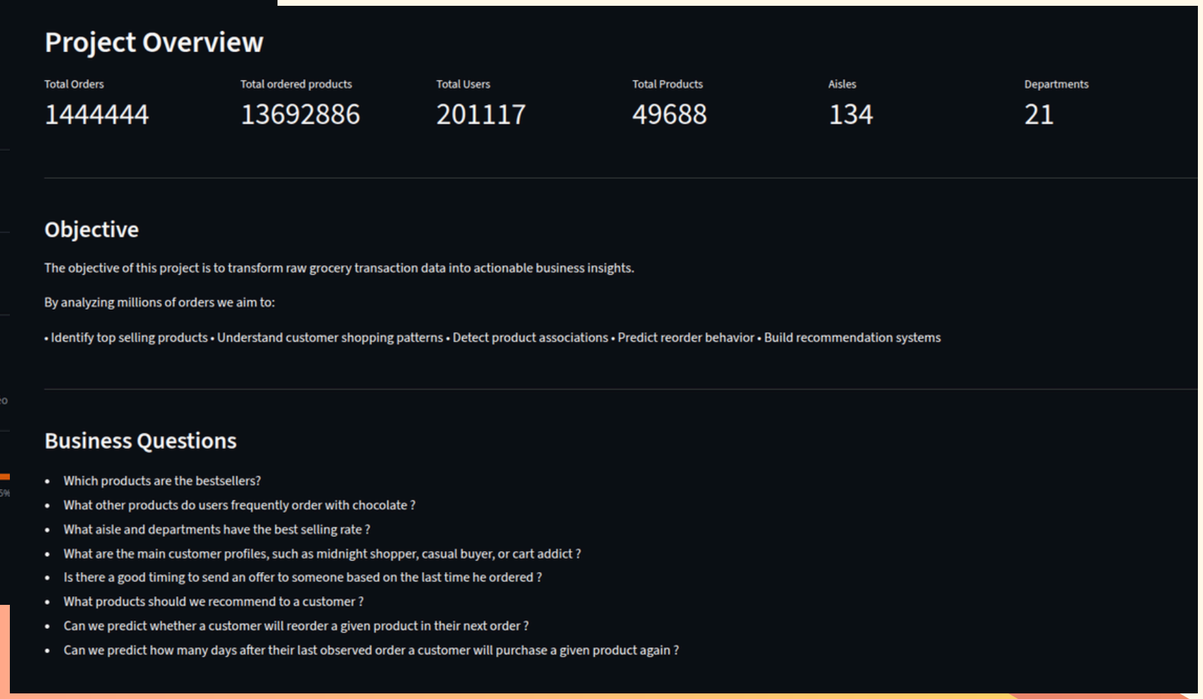


Les bonus réalisés

- Réalisation d’un dashboard intéractif avec Streamlit
- Prédictions basées sur une reconstitution de la temporalité
- Création de features engineering avancées
- Entraînement et test d’un model de type Gradient Boosting avec des features avancées
- Aggregation multi-level combinant produit, utilisateur et historique.
- Pipeline comprenant deux modèles afin d’en tirer de meilleurs résultats

Thank You


EpitechBachelorPromo2028 / B-DAT-201-PAR-2-1-nextbuy-20
Code
Issues 1
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
B-DAT-201-PAR-2-1-nextbuy-20 Private
Watch 0
Fork 0
Star 0
main
3 Branches
0 Tags
Go to file
Add file
Code
MXXR-Fivem Merge pull request #26 from EpitechBachelorPromo2028/theo
ab32889 · 4 days ago
60 Commits
assets update output cell 4 days ago
datasets feat(data analyse) : Add csv files 4 days ago
models feat(data-analyse) : Starting unit test, cleaning of the repo... 4 days ago
pages update output cell 4 days ago
tests feat(data cleaning) : Ruff config / ruff fix 4 days ago
.gitignore chore(dashboard) clean 4 days ago
Dashboard.py feat(data cleaning) : Ruff config / ruff fix 4 days ago
README.md feat(data analyse) : Merge the three datasets in notebook... 4 days ago
data_loader.py update output cell 4 days ago
notebook.ipynb update output cell 4 days ago
pyproject.toml feat(data analyse) : Merge the three datasets in notebook... 4 days ago
requirements.txt feat(data-analyse) : Starting unit test, cleaning of the repo... 4 days ago
README
NextBuy, From raw data to smart decisions.
NextBuy is the result of a business research analysing the habits and tendencies in the data of a supermarket. Using data analysis and machine learning, it present some meaningful insight on the state of the market and some ways to optimize marketing and promotion.
License
About
No description, website, or topics provided.
Readme
Activity
Custom properties
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 2
Jules-Epitech
MXXR-Fivem Théo
Languages
Jupyter Notebook 96.5%
Python 3.5%


Project Overview
Total Orders 1444444
Total ordered products 13692886
Total Users 201117
Total Products 49688
Aisles 134
Departments 21
Objective
The objective of this project is to transform raw grocery transaction data into actionable business insights.
By analyzing millions of orders we aim to:
• Identify top selling products • Understand customer shopping patterns • Detect product associations • Predict reorder behavior • Build recommendation systems
Business Questions
Which products are the bestsellers?
What other products do users frequently order with chocolate ?
What aisle and departments have the best selling rate ?
What are the main customer profiles, such as midnight shopper, casual buyer, or cart addict ?
Is there a good timing to send an offer to someone based on the last time he ordered ?
What products should we recommend to a customer ?
Can we predict whether a customer will reorder a given product in their next order ?
Can we predict how many days after their last observed order a customer will purchase a given product again ?